Beyond Accuracy: Stability-Oriented Learning for TCRI Credit Risk Prediction

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Outline

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2. Introduction
3. Data
4. Benchmark
5. Study plan
6. Results

Motivation

- Credit risk prediction is crucial for financial stability
- Taiwan's Standard: TCRI (Taiwan Corporate Credit Risk Index).
    - Covers 90% of listed companies
    - Key metric for lending decisions
- The Problem: Existing TCRI studies lack comprehensive predictive modeling

Introduction

- Past Approaches:
  - Financial Ratios (e.g., Altman Z-score).
  - Machine Learning (SVM, Random Forest, Neural Networks).
- What is Missing?
  - Temporal Leakage: Ignoring the time factor in data splits
  - Calibration: Probabilities are not "real" risk scores
  - Interpretability: "Black box" models are hard to trust

Introduction

- Key Methodologies:
  - Time-Aware Split: Prevent information leakage
  - Probabilistic Calibration: Interpretable risk scores (Platt Rescale)
  - Interpretability: SHAP values for feature importance
- Goal: A transparent, stable, and practical decision-support tool

Data

- Dependent variable: TCRI score
- Independent variable: the most significant financial ratios identified in Beaver and Altman
    - Working Capital / Total Assets
    - Retained Earnings / Total Assets
    - Cash Flow / Total Debt
    - Total Debt / Total Assets
    - Current Ratio

Data Preparation and Cleaning

- To ensure data quality, we addressed missing values and removed extreme outliers before analysis.

1. Missing Value Imputation
   - Create a missing value indicator for each independent variables
   - imputed using the median of firms with the same first two digits of the company code.
   - If the whole group are missing → use the overall sample median
2. Outlier Removal
   - Observations where any ratio exceeded three standard deviations from the mean were excluded
3. Take the logarithm

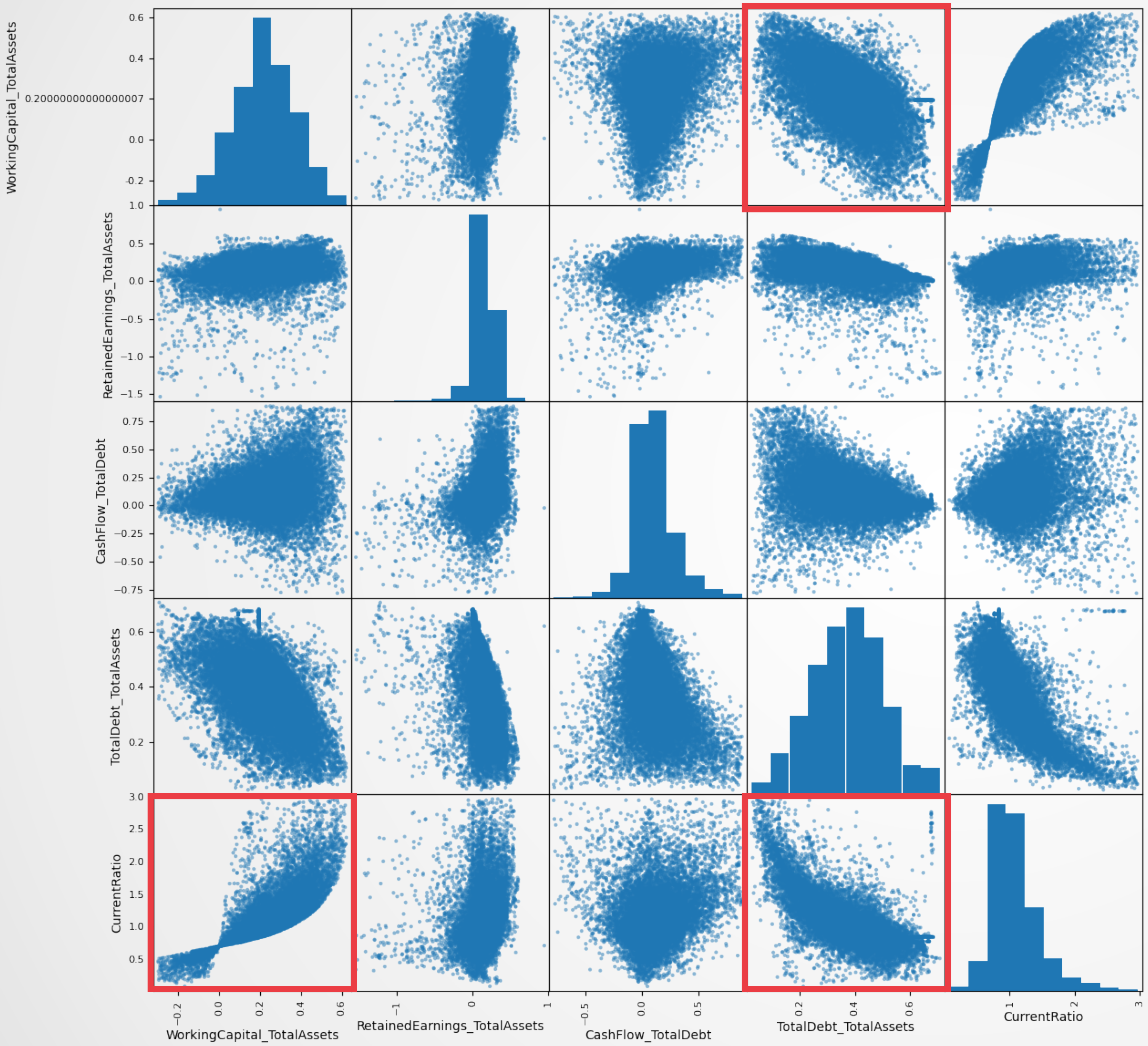
Key Descriptive Statistics

	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	3.13318	0	0.0964763	0	3.13318
mean	0.260084	0.111762	0.109787	0.457993	3.04299
std	0.205335	1.27573	0.49632	0.195864	11.1524
min	-0.742243	-82.0719	-18.1768	0.00303337	0.0588412
50%	0.251428	0.150736	0.080471	0.454693	1.79589
max	0.996967	1.58174	6.63415	1.05463	711.495
skewness	0.0955887	-44.8339	-10.2512	0.228632	29.3605
kurtosis	0.472559	2247.58	264.752	-0.219784	1243.48

Data cleaning

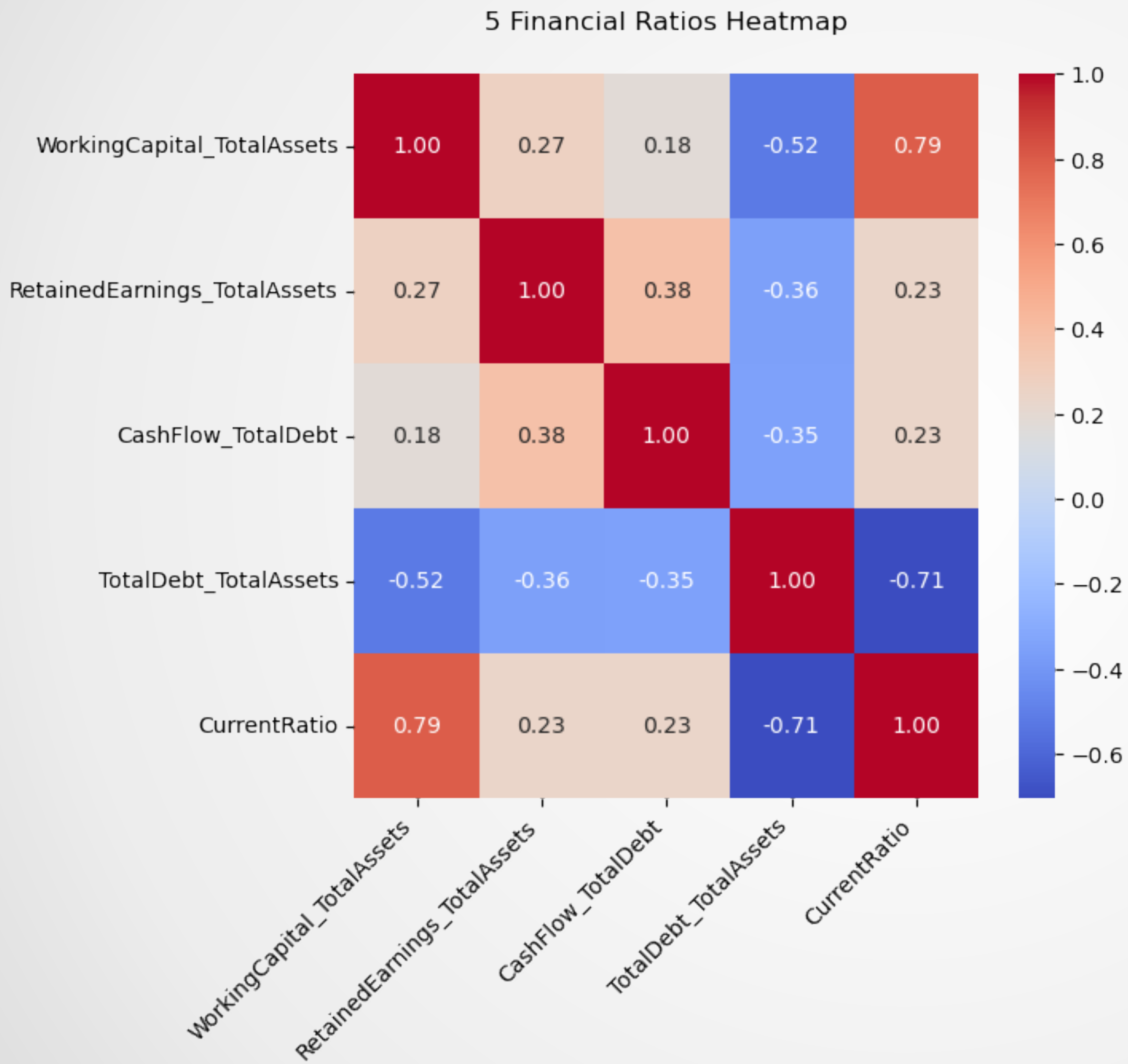
	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	0	0	0	0	0
mean	0.215023	0.135149	0.103381	0.373306	1.08464
std	0.15288	0.167935	0.186678	0.129052	0.380631
min	-0.298188	-1.53474	-0.78052	0.0258126	0.0741276
50%	0.215281	0.141115	0.0783935	0.378056	1.00802
max	0.623079	0.948465	0.885164	0.706987	2.96362
skewness	-0.242851	-2.41152	0.543679	-0.0259596	1.34961
kurtosis	0.0358163	15.6336	2.59051	-0.345129	2.99375

Data exploration



- Strong Linear Patterns:
  - TotalDebt_TotalAssets and WorkingCapital_TotalAssets (negative slopes).
- Non-linear Relationships:
  - CurrentRatio vs WorkingCapital_TotalAssets,
  - CurrentRatio vs TotalDebt_TotalAssets.

Data exploration

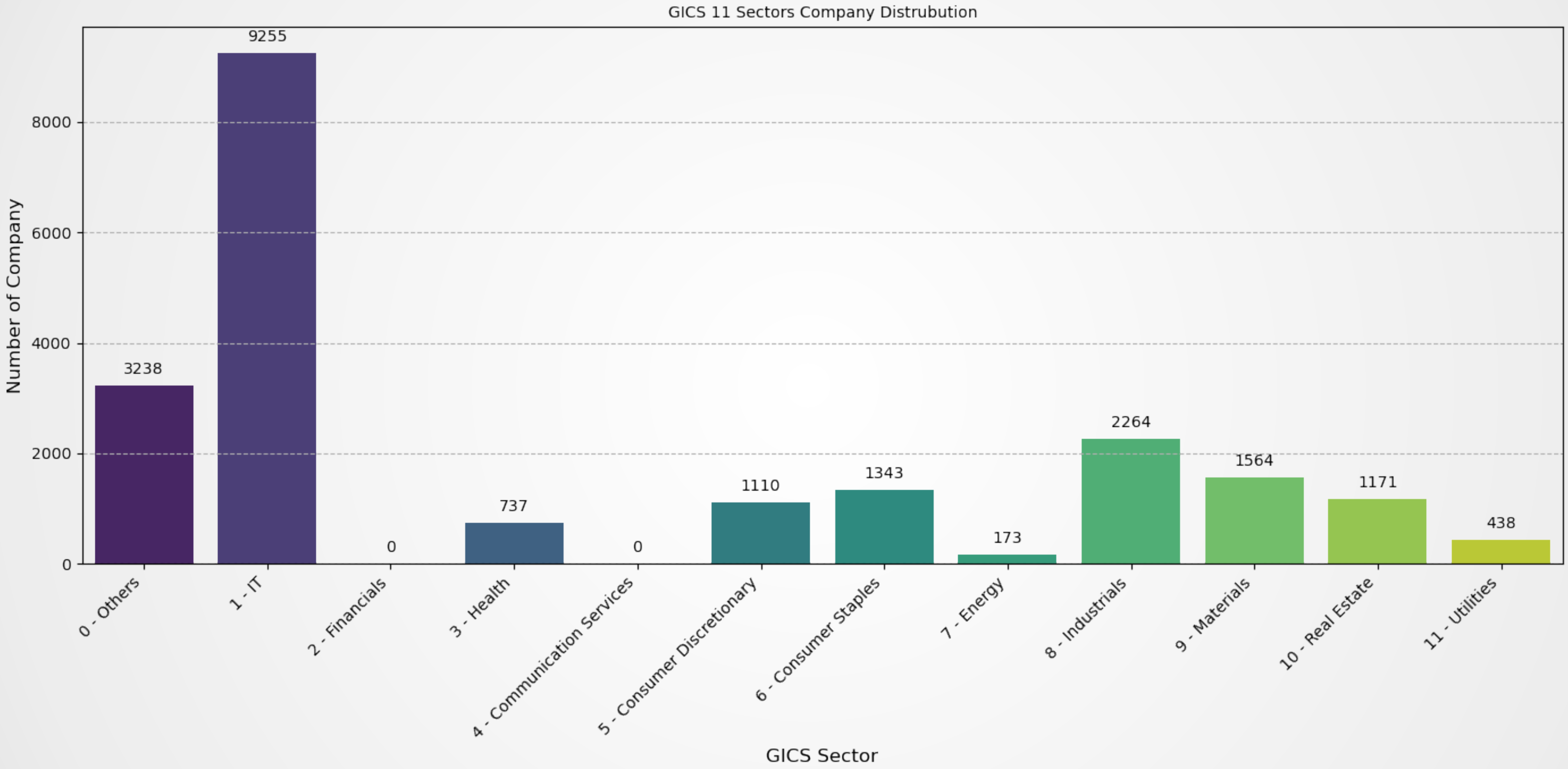


- High Positive Correlation:
  - WorkingCapital_TotalAssets and CurrentRatio → 0.79
  - companies with more working capital tend to have better liquidity.
- High Negative Correlation:
  - TotalDebt_TotalAssets and CurrentRatio → -0.71
  - Companies with higher debt levels typically have lower liquidity ratios.

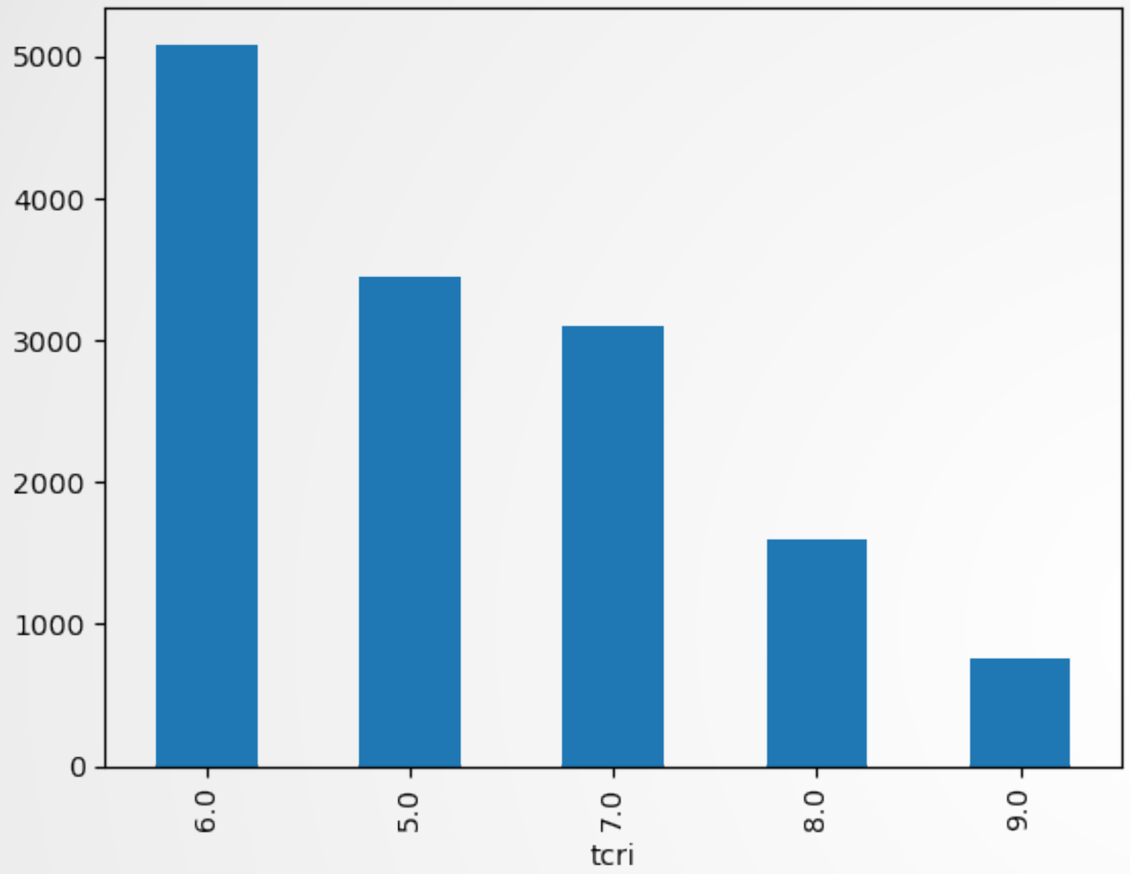
Data exploration

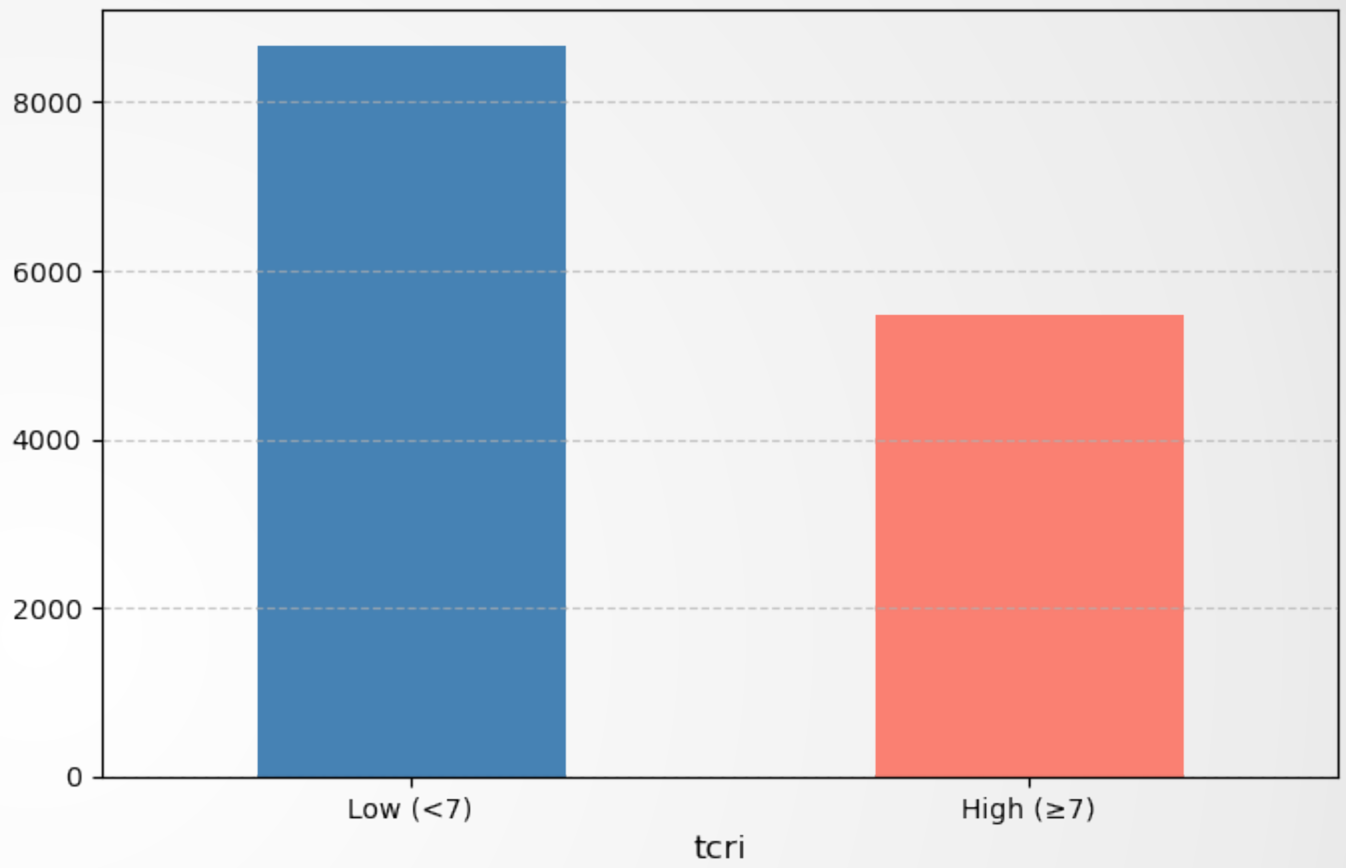
dummy variable	GICS Industry Sector	dummy variable	GICS Industry Sector
1	Information Technology	6	Consumer Staples
2	Financials	7	Energy
3	Health Care	8	Industrials
4	Communication Services	9	Materials
5	Consumer Discretionary	10	Real Estate
		11	Utilities

Data exploration



Data exploration (TCRI)





- The grouped distribution is not severely imbalanced, though some moderate skew remains.

Data exploration

- Ratios provide **complementary signals**—moderate overlaps between leverage and liquidity, but largely **distinct information profiles**, supporting their joint use in **credit rating modeling** and **SHAP analysis**.

Benchmark

- Dataset (2014-2024):
  - TEJ financial ratio
  - TCRI ratings
- Use **t-year** financial ratios to predict **t+1** high-risk status (binary)
  - High risk (1): tcri >= 7
  - Low risk (0): tcri < 7

Benchmark

- Sample Size / Positive Rate

		n	pos	pos_rate_%
	split			
2015–2022	Train	4624	1602	34.65
2023	Valid	616	221	35.88
2024	Test	619	218	35.22

- Logistic Regression

Test Performance

- Platt rescale
  - Probabilities more reliable
  - Operating-point trade-off
    - Precision
    - Recall
    - F1

	f1	precision	recall	brier
raw	0.7017	0.6125	0.8212	0.1556
platt	0.7043	0.6788	0.7318	0.1416

Forecast Result

- Captures **~82%** of truly high-risk
- About **60%** flagged firms are truly high-risk

	n	tp	fp	fn	tn	precision	recall	accuracy
forecast_year								
2024	619	147	93	32	347	0.6125	0.821229	0.798061

Flow chart

Raw Data

1. Missing Value Imputation
2. Outlier Removal
3. Take the Logarithm
4. Data Exploration
   - industry categorization
     - Data Exploration

X: Financial ratio

y: { 0: low risk
1: high risk

Train: 2014-2022
Valid: 2023
Test: 2024

Validation:
1. Platt Rescale

ML ways:
1. baseline
   Logistic Regression
2. Non-Linear
   Random Forest
   XGBoost
3. Non-linear & Non-Tree
   SVM （RBF kernel)
   Neural Network
4.quantile regression tree

Performance Metrics:
1. AUPRC
2.ROC-AUC

Calibration Metrics
1. Brier Score
2. ECE
3. Reliability Curves

Robustness Checks
1. different threshold definitions
2. Alternative dataset splits

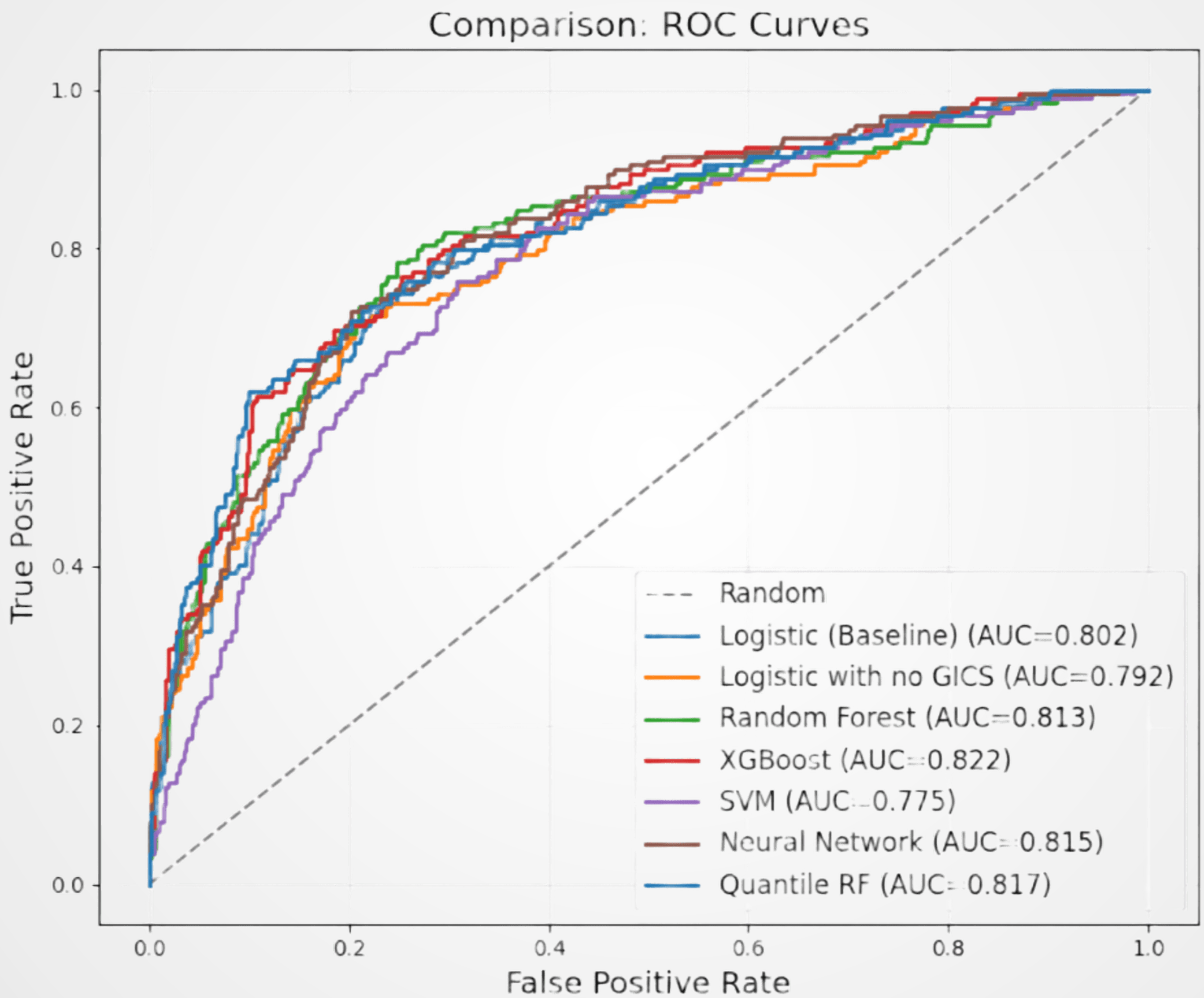
Interpretability
1.Feature importance
2. SHAP

Results

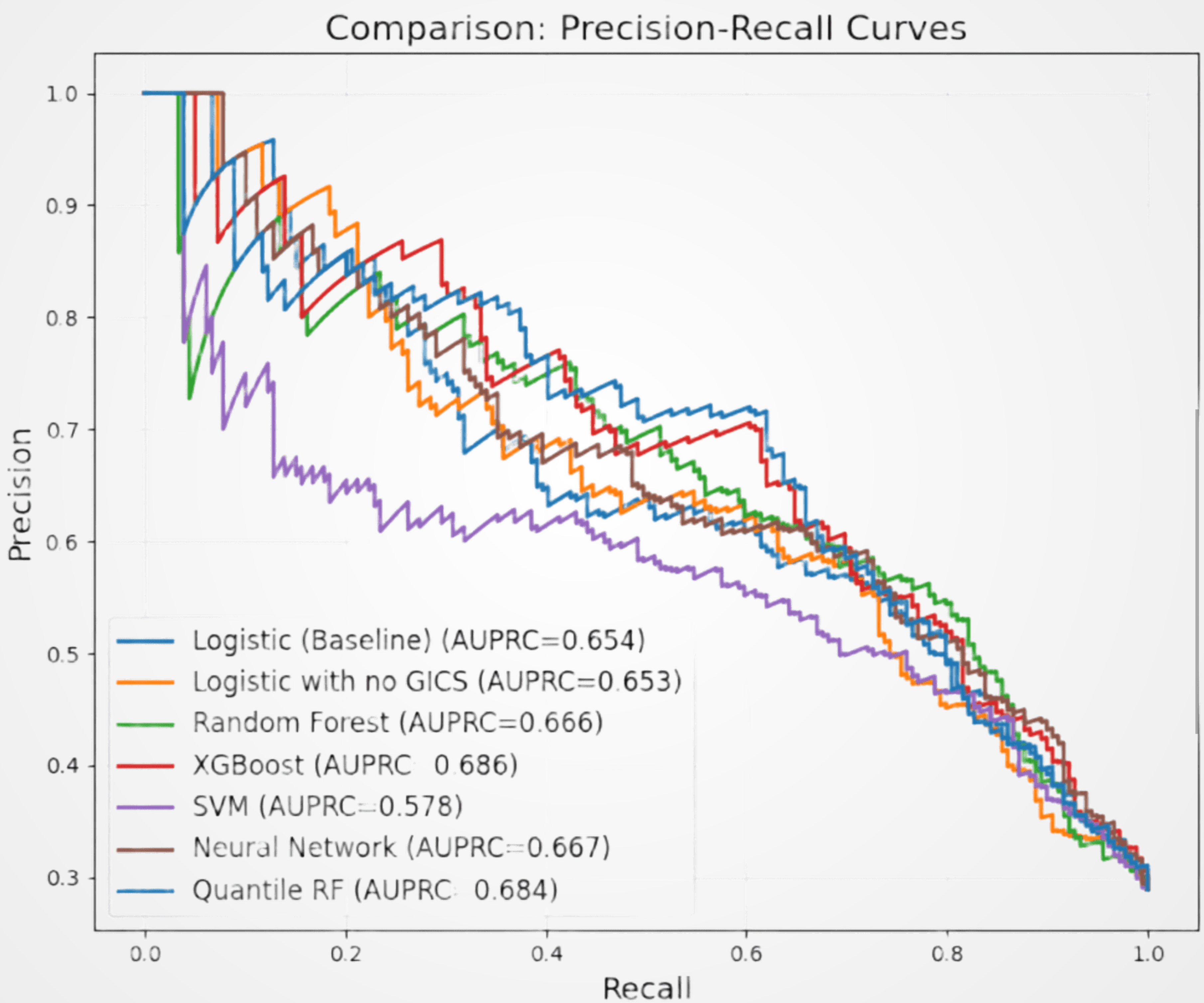
Model Name	AUC	AUPRC	Brier Score	ECE
Logistic Regression	0.8018	0.6542	0.1575	0.0633
Logistic Regression (no GICS)	0.7920	0.6533	0.1589	0.0681
Random Forest	0.8133	0.6662	0.1503	**0.0571**
XGBoost	**0.8222**	**0.6864**	0.1499	0.0638
SVM	0.7751	0.5780	0.1701	0.0583
Neural Network	0.8154	0.6674	0.1553	0.0634
Quantile Random Forest	0.8173	0.6840	**0.1490**	0.0715

- XGBoost has the highest AUC (0.8222) and AUPRC (0.6864)
- baseline Logistic Regression outperforms the Logistic model without GICS
- Random Forest achieves the lowest ECE, implying it provides the most well-calibrated probabilities

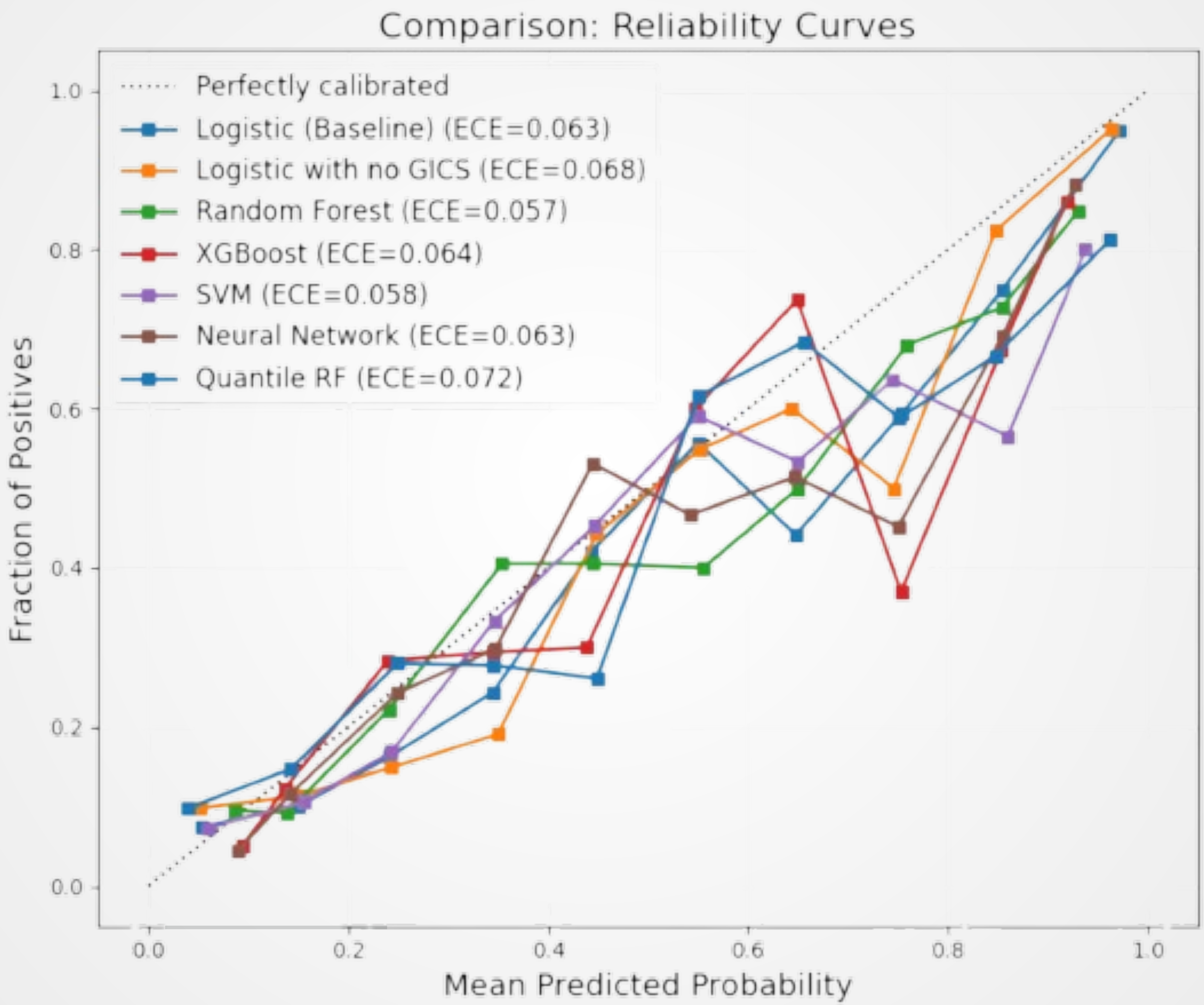
Results


Comparison: ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random
Logistic (Baseline) (AUC=0.802)
Logistic with no GICS (AUC=0.792)
Random Forest (AUC=0.813)
XGBoost (AUC=0.822)
SVM (AUC=0.775)
Neural Network (AUC=0.815)
Quantile RF (AUC=0.817)

Results


Comparison: Precision-Recall Curves
Precision
Recall
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8
1.0
Logistic (Baseline) (AUPRC=0.654)
Logistic with no GICS (AUPRC=0.653)
Random Forest (AUPRC=0.666)
XGBoost (AUPRC 0.686)
SVM (AUPRC=0.578)
Neural Network (AUPRC=0.667)
Quantile RF (AUPRC 0.684)

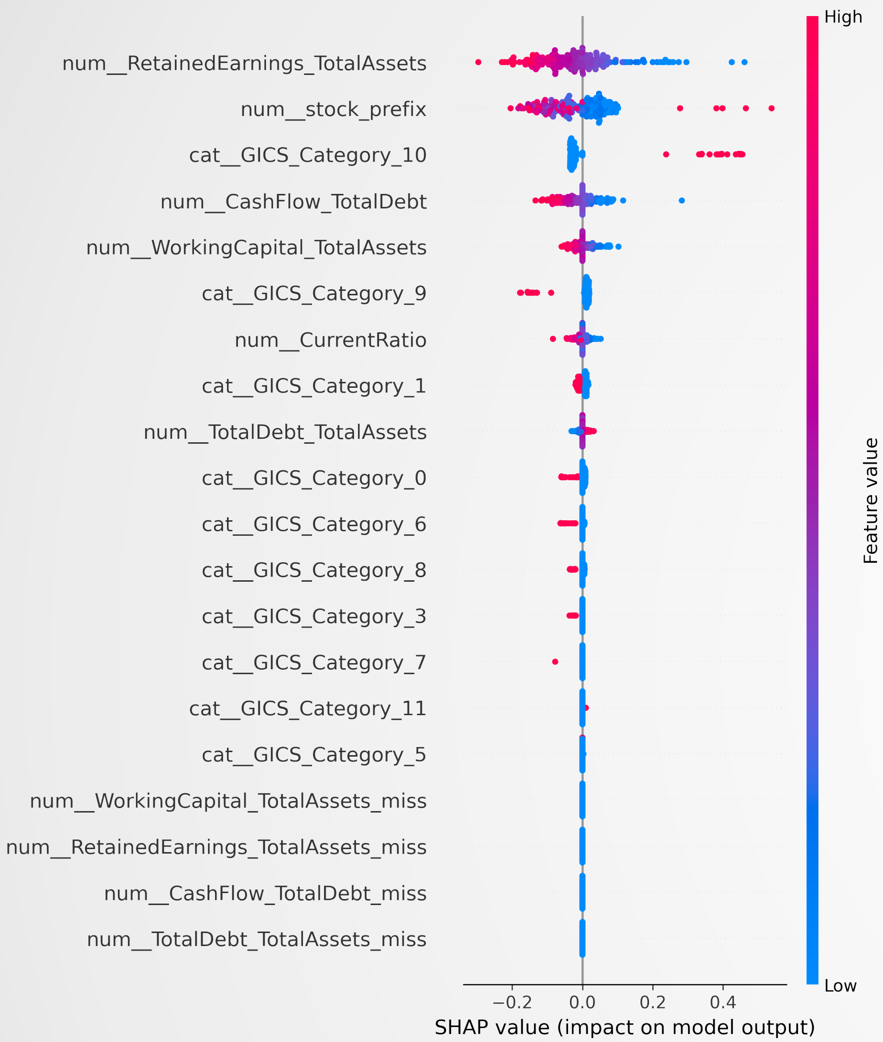
Results


Comparison: Reliability Curves
Perfectly calibrated
Logistic (Baseline) (ECE=0.063)
Logistic with no GICS (ECE=0.068)
Random Forest (ECE=0.057)
XGBoost (ECE=0.064)
SVM (ECE=0.058)
Neural Network (ECE=0.063)
Quantile RF (ECE=0.072)
Fraction of Positives
Mean Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0

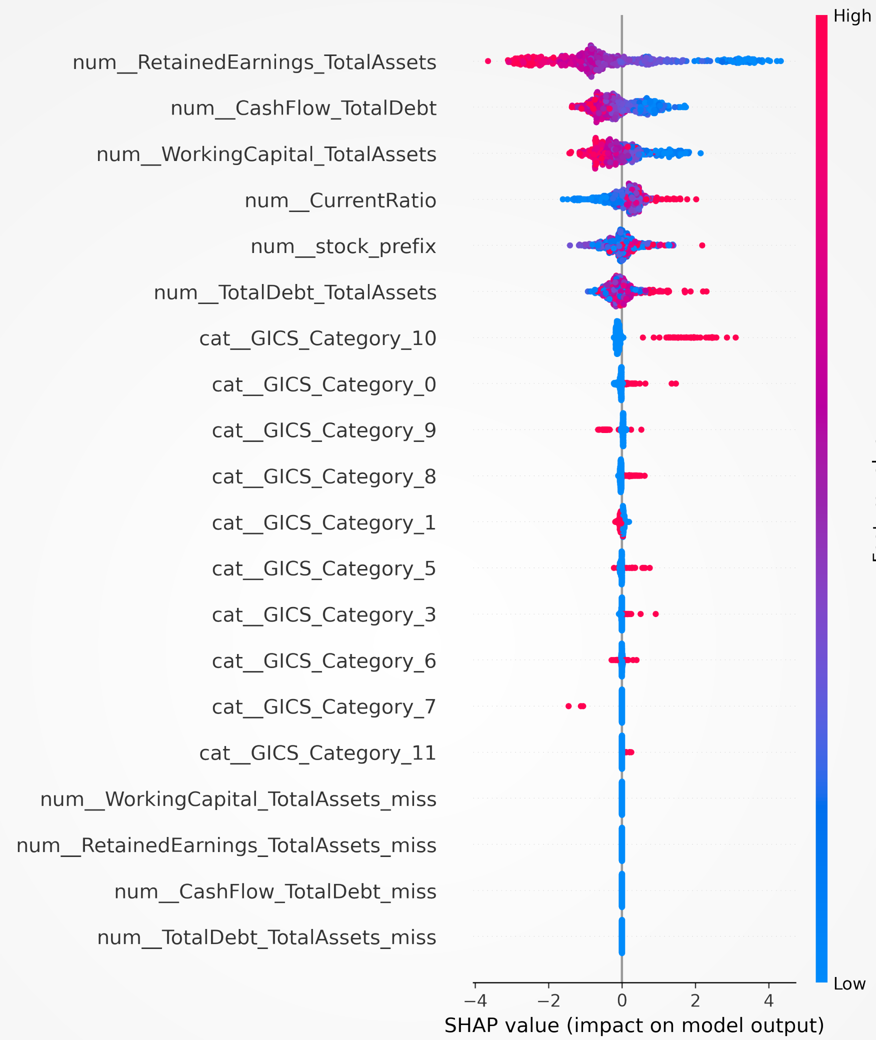
Results - Robustness Test

tau	calibration	pr_auc	roc_auc	f1	precision	recall	brier	ece
6	platt	0.748767	0.696226	0.729560	0.580000	0.983051	0.235200	0.140743
7	platt	0.654198	0.801778	0.597701	0.615385	0.581006	0.157525	0.063335
8	platt	0.577760	0.890327	0.444444	0.750000	0.315789	0.073633	0.031298
6	raw	0.748767	0.696226	0.737819	0.625984	0.898305	0.232563	0.119627
7	raw	0.654198	0.801778	0.612987	0.572816	0.659218	0.169858	0.112107
8	raw	0.577760	0.890327	0.573034	0.500000	0.671053	0.113915	0.166093

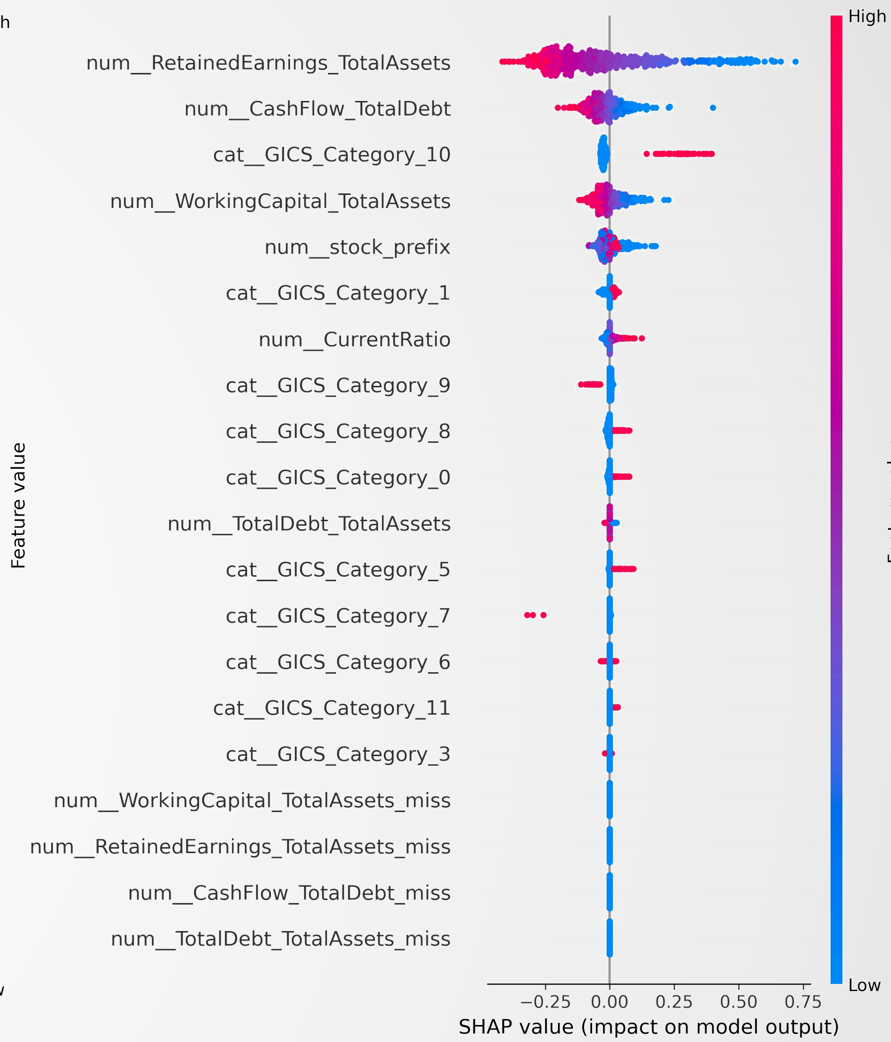
Results SHAP



Support Vector Machine

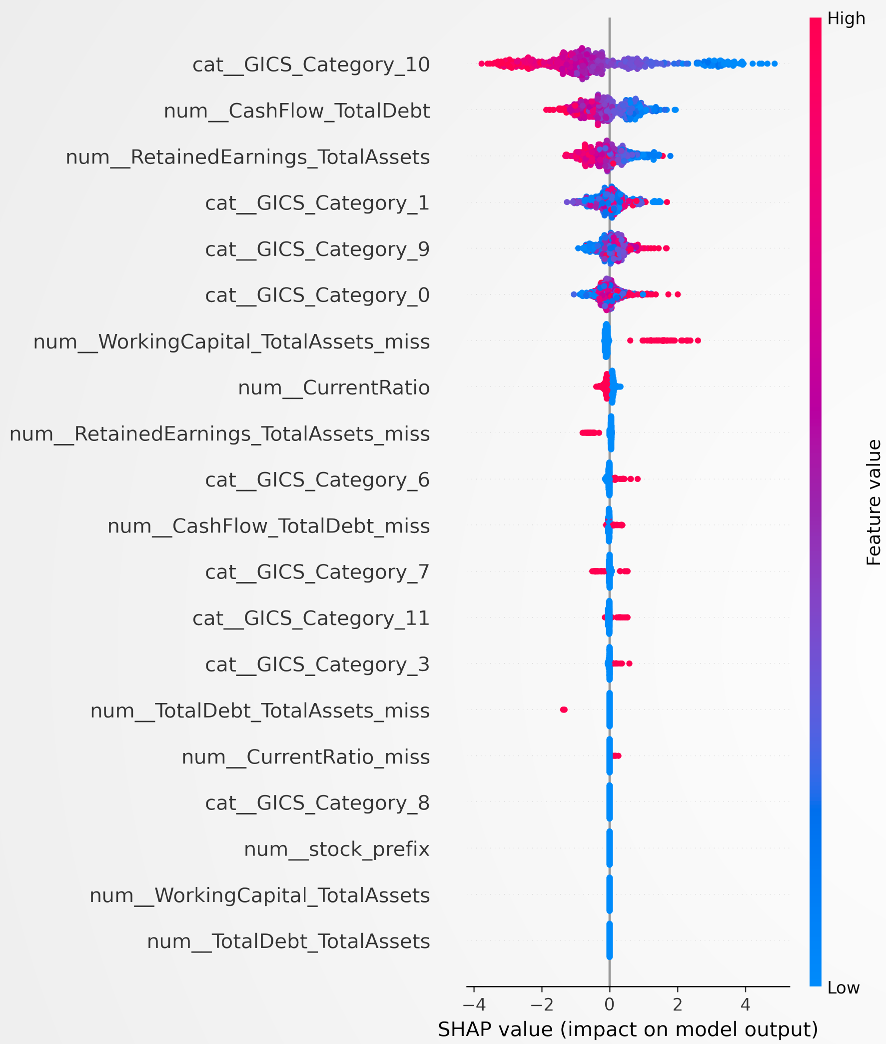


Quantile Regression Tree

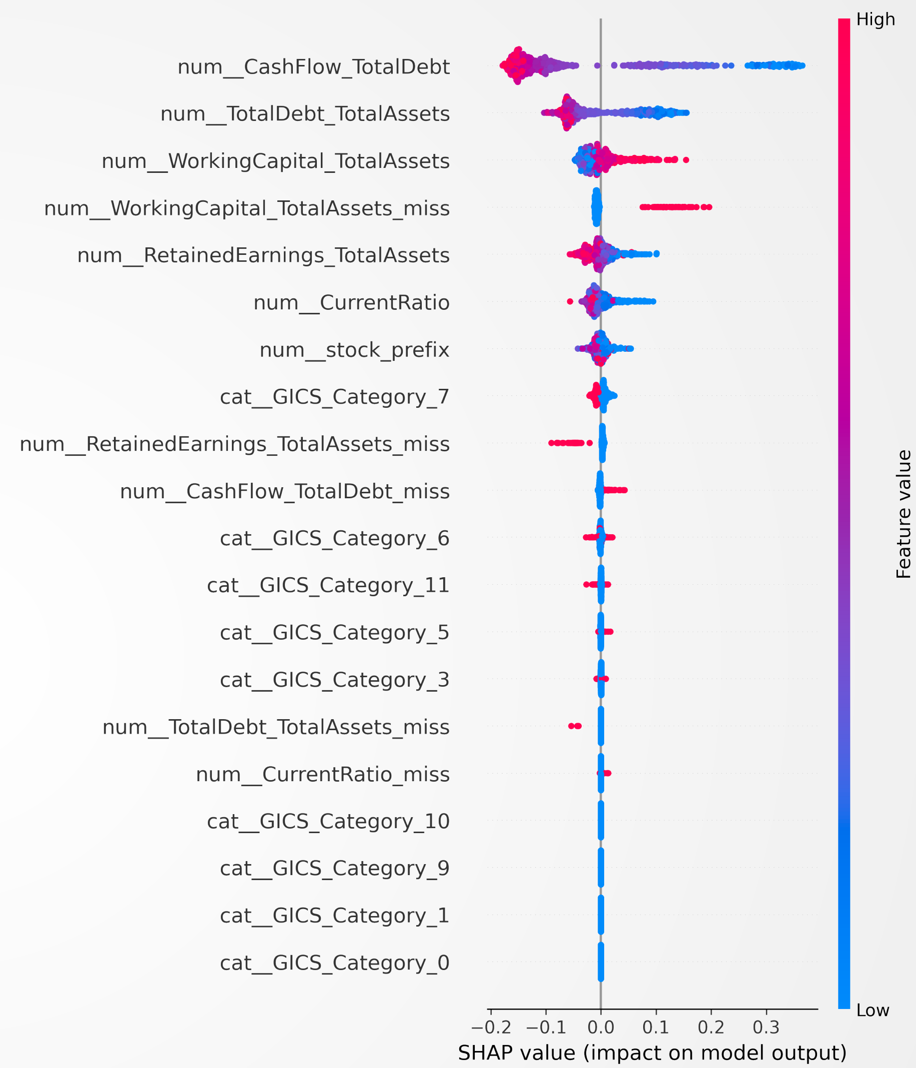


Neural Network

Results SHAP



XGBoost



Random Forest

Conclusion

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- XGBoost has best performance
- Industry is helpful in prediction
- Probability calibration is necessary when threshold is high.

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